Created using a free version of SMath Studio

Calculate atan(phase)

$$j := \sqrt{-1} \qquad S(phase) := \cos\left(\frac{phase}{2^{32}} \cdot 2 \cdot \pi\right) \cdot 2^{31} + j \cdot \sin\left(\frac{phase}{2^{32}} \cdot 2 \cdot \pi\right) \cdot 2^{31} \qquad N := 0$$

Atan2 approxymation

$$B(phase) := \mathrm{trunc}\left(\frac{\mathrm{trunc}\left(\left|\mathrm{Re}(S(phase))\right|\right)}{\mathrm{trunc}\left(\frac{\left|\mathrm{Im}(S(phase))\right|}{2^{16}}\right) + 0}\right)$$

$$A(phase) := \mathrm{trunc}\left(\frac{\mathrm{trunc}\left(\left|\mathrm{Im}(S(phase))\right|\right)}{\mathrm{trunc}\left(\frac{\left|\mathrm{Re}(S(phase))\right|}{2^{16}}\right) + 0}\right)$$

$$CorrXA(phase) := \left|phase - \left(2^{28}\right)\right| \qquad CorrXB(phase) := \left|phase - \left(2^{29} + 2^{28}\right)\right|$$

$$CorrA(phase) := \left(2^{26} - \frac{CorrXA(phase) \cdot CorrXA(phase)}{2^{30}}\right) \cdot 0.69$$

$$CorrB(phase) := \left(2^{26} - \frac{CorrXB(phase) \cdot CorrXB(phase)}{2^{30}}\right) \cdot 0.69$$

$$PhaseA(phase) := A(phase) \cdot 2^{13} + CorrA\left(A(phase) \cdot 2^{13}\right)$$

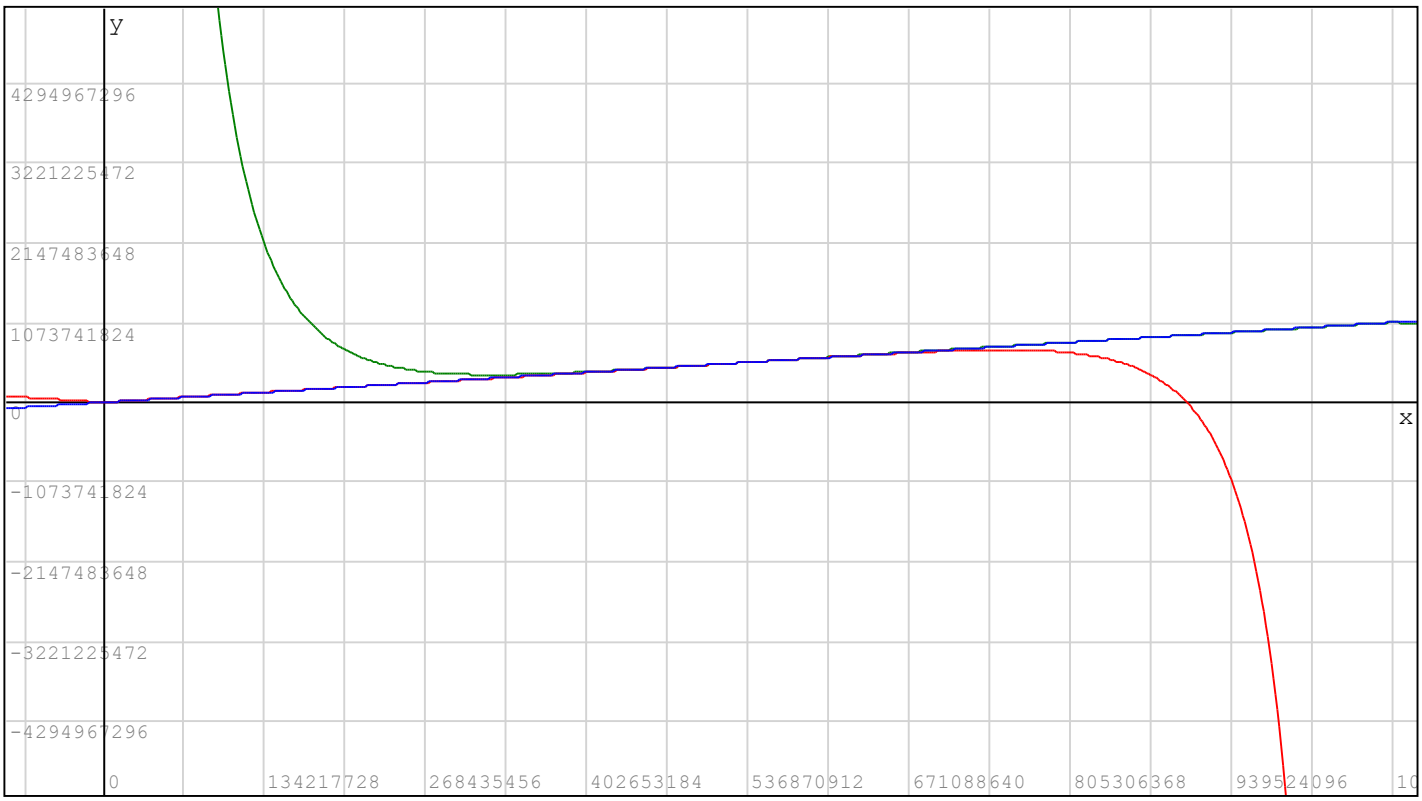
$$PhaseB(phase) := \left(2^{30} - B(phase) \cdot 2^{13}\right) - CorrB\left(2^{30} - B(phase) \cdot 2^{13}\right)$$

$$dPhaseA(phase) := phase - PhaseA(phase)$$

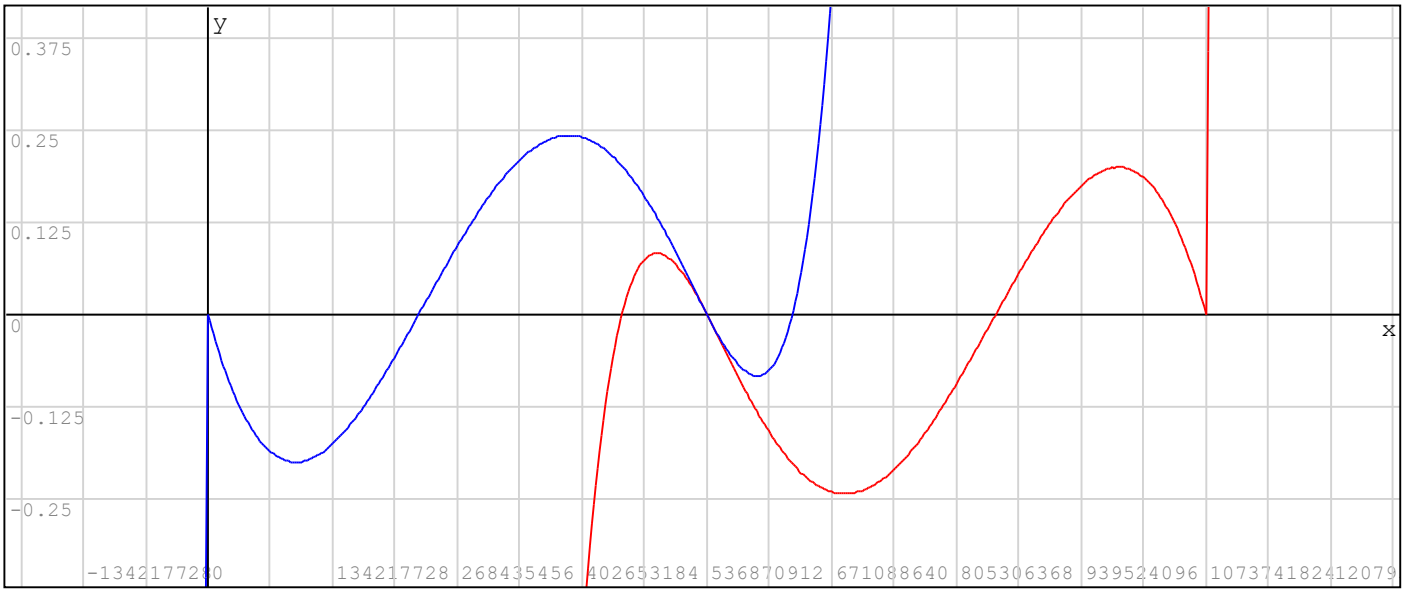
$$dPhaseB(phase) := phase - PhaseB(phase)$$

Not for commercial use

Created using a free version of SMath Studio




$$\begin{cases} x \\ PhaseA(x) \\ PhaseB(x) \end{cases}$$




$$\begin{cases} \frac{dPhaseA(x)}{2^{32}} \cdot 360 \\ \frac{dPhaseB(x)}{2^{32}} \cdot 360 \end{cases}$$

Not for commercial use